

Java Medium and Hard Level Problems

Input/Output Problems:

1. Write a program that reads an integer from the user and checks if it's a palindrome.
2. Create a program that takes a sentence from the user and prints each word on a new line.
3. Develop a program to read a list of integers from the user and display the maximum and minimum values.
4. Write a program that reads a text file and counts the frequency of each word.
5. Create a program to read a multi-line input from the user and store each line in a separate file.

Operators Problems:

1. Write a program that calculates the compound interest for given principal, rate, and time using arithmetic operators.
2. Create a program to check if a number is a power of 2 using bitwise operators.
3. Develop a program that performs bitwise operations to toggle the case of each character in a string.
4. Write a program to evaluate a mathematical expression entered by the user using arithmetic operators.
5. Create a program to perform modular exponentiation using only bitwise and arithmetic operators.

Conditional Problems:

1. Write a program that checks if a year is a leap year without using logical operators.
2. Create a program to find the largest of three numbers using ternary operators.
3. Develop a program to classify a triangle as scalene, isosceles, or equilateral based on side lengths entered by the user.
4. Write a program that takes an age as input and categorizes it as child, teenager, adult, or senior.
5. Create a program to determine if a given string is a valid IP address.

Looping Problems:

1. Write a program to find the factorial of a number using a for loop.
2. Create a program to print the Fibonacci sequence up to a given number using a while loop.
3. Develop a program to find the sum of all prime numbers between 1 and 100 using a for loop.
4. Write a program to count the number of vowels in a given string using a while loop.
5. Create a program to print a pattern of numbers in a pyramid shape using nested loops.

Nested Looping Problems:

1. Write a program to print Pascal's Triangle up to a given number of rows.
2. Create a program to print the multiplication tables from 1 to 10 using nested loops.
3. Develop a program to generate a pattern of a hollow square using asterisks with nested loops.
4. Write a program to display a diamond shape pattern with numbers using nested loops.
5. Create a program to print all possible combinations of a given set of numbers using nested loops.

Type Casting Problems:

1. Write a program to convert a double value to int and display the loss of precision.
2. Create a program to read a floating-point number and convert it to a string and integer format.
3. Develop a program that accepts a character and displays its ASCII value by typecasting.
4. Write a program that converts a string to a float and rounds it to two decimal places.
5. Create a program to convert an integer to binary, octal, and hexadecimal formats.